

Joint Assessment of Six Unifying-Object Audits

The κ_V coherence problem — full claim-level verification of six independent repair proposals, eight audit errors corrected, and one synthesized unification architecture assembled from the six.

Verification & Synthesis Unit

Manuscript lineage v12–v21 · frozen baseline v21 (commit ddbb384) · audits:
DeepSeek, GLM, GPT, Kimi, Muse, Opus

118 checkable claims verified against the frozen source; 111 confirmed, 8 corrected or stale. Layered synthesis: notation protocol, factored curvature datum, provable energy-level transfers, and three computable curvature measurements (second differences, plaquette, double-knockout epistasis). Manuscript untouched — all repairs specified for journal_manuscript_v2.

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PART I

Executive Summary and the Six Audits

The repository folder *external_audits/unifying object/* contains six new audits, 1,058 lines in total, all responding to one question: what would it take to make the manuscript a single coherent article, given that three disjoint mathematical objects currently share the name κ_V . The six authors arrive by different routes, propose materially different central objects, and yet converge to a remarkable degree on both the diagnosis and the shape of the repair. This assessment verifies every checkable claim against the frozen v21 manuscript (10,830 LaTeX lines, 134 PDF pages, commit ddbb384), corrects the audit errors it finds, completes the weaker suggestions, and synthesizes the six proposals into one architecture that none of them states alone.

6 audits verified line-by-line	1,058 audit lines read in full	38 claim-level verdicts in Part II	8 audit errors corrected in Part III	4 synthesis layers in Part V
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What all six agree on

Every audit accepts, and this assessment independently re-verified, the core factual basis: (i) the abstract and Section 4 make their main proposition about a geometric curvature built from a Bregman divergence of a smooth surrogate of an algorithmic rate-distortion distance (Proposition 4.4); (ii) the empirical flagship studies E24 to E27 correlate a per-gene statistic that is a squared flux displacement along a glucose-decline trajectory, $\kappa_V(r,t) = (v_r(t) - v_r(T1))^2$, taken unchanged from the E22 artifacts; and (iii) Definition 3.21 defines an indicator-weighted single-knockout flux-rerouting sum. These are three objects with different types, different units, and no theorem connecting them. GLM calls this "the single deepest coherence defect"; Opus calls the experimental object "the deficit function itself, not the curvature"; Muse states flatly that the flagship "does not test the abstract's proposition" unless a unification theorem is proven; Kimi warns that the strong form of that theorem is "probably not provable within this manuscript's scope". All six are right, and the line-level evidence is in Part II.

What the six propose, in one table

ID	Audit (model)	Central proposal	Character	Role in synthesis
U1	DeepSeek recommendations (188 lines)	Editorial reconstruction as a formal journal article; one named central object (unify or rename); Methods-not-change-log; all contradictions reconciled; "repair and demote over remove"	Editorial / structural	Style and architecture backbone; deletion-demotion policy
U2	GLM (194 lines)	Five-thread ontology (G/A/C/D/B); definitions D1-D6; seven theorem repairs T1-T7 with effort estimates; closure - geometry - computability image; explicit deletion list	Comprehensive / strategic	Structural backbone; T2 jump-cocycle and T7 mpLP bridge are the new mathematics
U3	GPT "sol" (351 lines)	Typed realization datum (B,E,pi,s,Y,a,phi,W,m,N); realized perturbation divergence Delta_R; kappa_geo as a separate curvature operator; four-condition plaquette estimator; biomass deficit is not a metric	Type-theoretic / careful	Formalization backbone; the plaquette design and the transport caveat
U4	Kimi k3max (68 lines)	Weak formal unification: one slot-family kappa[d, c, Lambda]; three instantiations; relation theorems R1-R4 with labeled epistemic status; knockout-simplex embedding	Constructive / mathematical	Constructive backbone; R1/R3 are the first provable flagship-abstract link

ID	Audit (model)	Central proposal	Character	Role in synthesis
U5	Muse spark 1.2 (90 lines)	Canonical κ_V^{geom} (Prop 4.4 form); rename the other two objects; Theorem U (conditional unification) or honest disambiguation table; 5-item migration checklist	Drop-in / tactical	Tactical backbone; the migration checklist is the implementation protocol
U6	Opus 5 (167 lines)	Curvature datum (X, A, Φ); central object = second mixed difference of the deficit; FBA = epistasis; E24-E27 measures Φ itself; Route C1 second differences (cheap) and Route C2 double perturbations	Factored / empirical	Empirical backbone; the only audit that converts the theory into new computations

Table 1. The six third-wave audits. Lines and file names from external_audits/unifying object/; characterizations are this assessment's.

Verification method and baseline

The same five-layer protocol as the second-wave assessment was used, now against the v21 source. Layer 1: line-level reading of scripts/journal_manuscript.tex (10,830 lines) with rendered-number extraction from the compiled PDF so that "Proposition 4.4" or "Remark 20.2" resolve to actual statements. Layer 2: full-text scans of the compiled PDF for counts and framings (for example "the user" appears 6 times case-sensitively, "Qwen" 34 times, the commit hash 07e6d85 five times, COLOMBOS three times). Layer 3: independent re-derivation of every mathematical claim that could be checked by hand - the softmax gradient identity behind GLM's Lemma 4.10 repair, the Fisher-Rao radius computation behind its Proposition 3.16 finding, the Stokes-substitution arithmetic behind Corollary 4.14, the $0.92^6 \times 1.15 = 0.697$ product behind Theorem 8.2. Layer 4: the deposited result files (CSV and JSON artifacts) for the Lévy exponent confidence interval and the E24/E25 numbers. Layer 5: code reading of the analysis scripts where a claim was about behavior rather than text. One baseline note matters for interpreting the audits: they reviewed the 131-page v20 PDF (GLM cites "the current 131 pages"); v21 (134 pages, commit ddbb384) had already executed the second-wave P0-mechanical round. Every verdict below is therefore re-verified against v21, and items the audits list that v21 already fixed are marked RESOLVED-IN-v21 rather than counted as audit errors.

PART II

Verification Ledger

Each row records an audit claim, its source audit, the verdict of this assessment, and the line-level evidence. Line numbers prefixed L refer to the v21 PDF text layer; "tex" refers to the LaTeX source. Verdicts: VERIFIED (claim confirmed against v21), CORRECTED (claim right in substance, wrong in a detail that matters for implementation), REFUTED (claim false), CONDITIONAL (claim holds only under a stated hypothesis), RESOLVED-IN-v21 (true of v20, already fixed in v21).

II.1 The central finding: three objects named κ_V

Claim	Audit	Verdict	Evidence in the frozen v21 source
Object (a): the geometric κ_V is a Bregman divergence of the smooth surrogate, positive part, sup over unit bivectors, per-area units	all six	VERIFIED	Prop 4.4 (L851): $h_{\alpha} = D_{\Phi}(\text{dist}_D(x), \text{dist}_D(x_0))$; L886-887: "the wedge-norm constraint makes κ_V dimensionally a per-area quantity"
Object (b): Def 3.21 is an indicator-weighted single-knockout squared-flux sum	all six	VERIFIED	Def 3.21 (L774): $\kappa_V(g) = 1[\Delta_b > \tau b_{\text{wt}}] \times \sum \text{over } dR(g) \text{ of } (v_r(\text{KO}) - v_r(\text{WT}))^2$; title still reads "v10 main definition, indicator-weighted"
Object (c): E10/E22 time-course statistic is $(v_r(t) - v_r(T1))^2$	U2 U3 U5 U6	VERIFIED	Sec 19.17 (L6093): "reaction-level $\kappa_V(r, t) = (v_r(t) - v_r(T1))^2$ with the identical E10/v13 definition"

Claim	Audit	Verdict	Evidence in the frozen v21 source
The empirical flagship E24-E27 uses object (c), not (a) or (b)	U2 U3 U5 U6	VERIFIED	Sec 19.19 (L6293): "The E22 panel's per-gene kappa_V values (baseline glucose-decline trajectory, unchanged from E22) are tested against M3D"; E26/E27 pair the same panel with protein and Schmidt-transcript metrics (L6513, L6588)
The abstract's main proposition concerns object (a) - hence the theoretical object never touches the empirical object	U2 U4 U5 U6	VERIFIED	Intro (L100-101): "the derivation of kappa_V from a Bregman divergence evaluated at dist_D (Proposition 4.4), supplying ... the per-trajectory falsification protocol"
Def 6.1 cites "the kappa_V of Definition 2.1" while Def 2.1 defines D_V and disclaims curvature	U5 U6	VERIFIED	Def 2.1 title (L204): "Viability depth functional D_V; distinct from curvature"; body: "it is not a curvature 2-form"; Def 6.1 (L1192) cites both anyway

Table 2. The three-objects finding and its evidentiary basis.

The flagship-uses-object-(c) claim required its own verification because it is the load-bearing fact of the whole wave: if E24 had used the FBA knockout statistic (object b), the unification problem would be a different one. The E24 design text is explicit, however, and so is E25 (GSE64021) and the E26/E27 pairing: all four studies correlate per-gene maxima of the trajectory statistic against transcript or protein response metrics. It follows that the manuscript's strongest empirical result - $r = +0.374$, $n = 433$, $p = 8.2 \times 10^{-16}$, partial $r = +0.251$ - is a statement about a squared displacement along one trajectory, while the abstract advertises a curvature contraction. Opus's phrasing is the sharpest and is confirmed: the experiments report a first-order object; the theory promises a second-order one.

II.2 The load-bearing theorem defects

Claim (theorem-level)	Audit	Verdict	Evidence and independent check
Corollary 4.14's proof substitutes $F = \text{kappa_V}$ and $\text{area} = \pi a^2$ into Stokes, "giving $H_{\text{geo}} = \pi a^2$ " - dimensionally incoherent ($\pi a^4 / V_{\text{max}}$ unless $a^2 = V_{\text{max}}$)	U2 U4 U5 U6	VERIFIED	Cor 4.14 proof (L1093-1097) verbatim: "substituting $F = \text{kappa_V}$ and $\text{area}(\text{gamma_a}) = \pi a^2$ gives $H_{\text{geo}} = \pi a^2$ ". Re-derived: Stokes gives $H = \text{kappa_V} \times \pi a^2 = (a^2/V_{\text{max}}) \times \pi a^2$; equality to πa^2 is a coincidence, not a theorem
Lemma 4.10's uniform bound is vacuous as τ approaches 0 (denominator bound decays exponentially)	U2 U4	VERIFIED	Proof (L980-992): denominator bounded below by $2^{-L/\tau} \times e^{-\beta \text{diam}(K)/\tau}$, which tends to 0; the claim "uniformly bounded over the family" is false as stated
GLM's repair identity: $\text{grad } r = \beta \times \sum \pi_j \text{grad } q_j$, hence $ \text{grad } r $ is of the order $2 \beta L_d (\text{diam} + D)$, uniform in τ	U2	VERIFIED	Independently re-derived: $r = -\tau \log \sum w_j e^{-\beta q_j / \tau}$, so $\text{grad } r = \beta \times \sum \pi_j \text{grad } q_j$ - the outer $-\tau$ cancels the $1/\tau$ in the exponent derivative; the τ -dependence disappears exactly as GLM claims. Constant note in Part III
Theorem 4.11(e) claims upper-semicomputability "as the sup of a computably enumerable family" while the family is indexed by a continuum	U2 U4 U5 U6	VERIFIED	Thm 4.11(e) (L1003-1010): $\text{kappa_alg} = (1/V_{\text{max}}) \sup \text{ over } a \text{ in } (0, a^*]$ and unit bivectors of $E(x; F(u, v))$; both (τ, β, D) and $(a, \text{bivector})$ are continua. Additional finding: the direction sup is a second continuum the audits did not flag explicitly - see Part V for the closure
Remark 20.2 asserts $\text{kappa_V} \leq E$ - a derivative-ratio bounded by a function value (unit error)	U2 U4 U5 U6	VERIFIED	Remark 20.2 (L6743-6744): " $\text{kappa_V}(\theta, x) = [-D h_{\alpha}] + / h_{\alpha}$... is therefore bounded above by the algorithmic envelope E ". kappa_V has per-area units, E is a value; the stated justification (Lipschitz + Clarke subdifferential) does not yield the comparison
E13 Theorem A: $\Phi(S) = \{r \text{ in } U : \dots\}$ is claimed "weakly contractive ($\Phi(S)$ contained in S)" - does not follow as displayed	U2 U4 U6	VERIFIED	Thm A (L5010-5014): membership is $r \text{ in } U$, not $r \text{ in } S$; $\Phi(S)$ contained in S fails until the one-line fix $r \text{ in } S$ is made (GLM D4)
E13 Theorem B: uniqueness "by Strachey-Reynolds parametricity" - parametricity constrains polymorphic terms, not functors	U2 U4 U6	VERIFIED	Thm B (L5038): "to R (uniqueness by Strachey-Reynolds parametricity restricted to typed polynomial optics)". No free theorem of this form exists; GLM and Opus independently recommend deletion, Kimi proposes adjoint-uniqueness restatement

Claim (theorem-level)	Audit	Verdict	Evidence and independent check
Corollary 17.3 is contentless as a classification ("identified by the univalence axiom with a single term in the HoTT universe U")	U2 U4 U6	VERIFIED	Cor 17.3 (L2697-2705) verbatim; the type of contractible groupoids is itself contractible, so the identification carries no classification content; the univalence invoke is decorative
Theorem 3.5 says a loop crosses a separating boundary "once" while the manuscript's own numerics give $n_{\text{cross}} = 2$ (rectangular and pentagon loops)	U2	VERIFIED	Thm 3.5 (L412): "of area epsilon2 crossing the boundary once"; verification scripts report (L448-453): "crossing the boundary twice ... $n_{\text{cross}} = 2$ "; a closed transverse loop crosses a separating boundary an even number of times
Theorem 3.14's tuple (Theta, pi, Δ^{n-1} , epsilon, Gamma) has drifted from Definition 3.1's (B, E, P, epsilon, Gamma)	U2 U4 U6	VERIFIED	Thm 3.14 (L616) vs Def 3.1 (SAVGS, Sec 3); both renderings confirmed in the v21 PDF text
Proposition 3.16's "unit sphere" is wrong by a factor of 2 (the Fisher-Rao simplex is radius-2 spherical under $\psi = 2 \sqrt{p}$)	U2	VERIFIED	Prop 3.16 (L672-677): " $\psi_a = 2 \sqrt{p_a}$ places the open simplex on the positive orthant of the unit sphere" together with $d_{\text{FR}} = 2 \arccos(\dots)$. Checked on the binary simplex: geodesic integral of $dt/\sqrt{t(1-t)}$ from 0 to 1 equals π , which is the radius-2 geodesic, not the unit-sphere geodesic $\pi/2$; the image of $2 \sqrt{p}$ has norm 2. GLM's radius-2 finding is correct
Remark 2.4's "the Fisher-Rao stabilizer is $\text{CO}(2) = \mathbb{R} + \text{semidirect } \text{SO}(2)$ " is a label error; the round-metric point stabilizer is $\text{O}(n-1)$	U2	VERIFIED	Remark 2.4 (L246-248) states $\text{CO}(2)$ flatly, while Remark 2.3 (L239-244) correctly derives when a $\text{CO}(r)$ reduction is justified (Fisher-Weyl policy structure with positive scale). The Lie-algebra content ($\text{so}(2)$ abelian at $n = 3$, $\text{so}(3)$ non-abelian at $n \geq 4$) survives with corrected labels
Proposition 19.7's Zeno lower bound $\tau \geq 1/(1 + \log_2(N_{\text{RAF}}))$ is manufactured (closure depth is not logarithmic)	U2	VERIFIED	Prop 19.7 (L4263-4272) states exactly that bound; chains can descend by one reaction per step, so stabilization in at most $ U $ iterations at $\text{O}(U \times R)$ cost is the true statement (GLM T4)
The fitted $3/2$ fatigue exponent framing ("verified to within 1.4%") overstates: the theory value 1.500 lies outside the bootstrap CI [1.471, 1.493]	U2	CORRECTED	The 1.4% framing appears at L130, L2025, L7246; the CI disclosure appears once (L2013-2017) with an honest bias explanation. GLM's "report that the fitted exponent lies outside the quoted CI" misstates the direction (it is the theory value that lies outside the fitted CI, and the manuscript does disclose it once) - the real defect is the inconsistent framing across sites, not a hidden CI

Table 3. Theorem-level defects, with this assessment's independent re-derivations where the claim was checkable by hand.

II.3 Type-level and structural defects

Claim	Audit	Verdict	Evidence
dist_D - defined on strings (Def 4.1) - is applied to RAF sets	U2 U5 U6	VERIFIED	$D_{\text{phi}}(\text{dist}_D(R), \text{dist}_D(R_0))$ appears at three sites: Table 1 optic O1 residual (L1232), the RAF-optic definition (L1291), and the Section 16 composite (L2398). Location sharpened relative to the audits' "Def 3.21" attribution
Def 2.5 makes the residual part of the optic OBJECT (triple (M, C, R)) while citing Riley, whose objects are pairs with existentially-quantified residuals	U2 U3 U4 U6	VERIFIED	Def 2.5 (L264-268): "objects triples (M, C, R) ... (Proposition 2.3 of [4])"; the citation does not match the structure it is cited for
The realization functor R is invoked six times and never defined; Remark 7.8 concedes the gap	U2 U3 U4 U6	VERIFIED	Remark 7.8 (L1357-1367): "The full functorial construction is open"; "a realization functor R from endo-optics to endomaps" is used without construction
Theorem 8.2 is proven for seven generic resnet-style blocks, not for the seven optics of Table 1	U2 U3 U6	VERIFIED	Sec 8.1 (L1406-1417): $f_i(x) = (1 - a_i) x + a_i \sum s_i \sigma(W_i x + b_i)$; the numbers never refer to the categorical objects; product $0.92^6 \times 1.15 = 0.6973$ verified arithmetically

Claim	Audit	Verdict	Evidence
"Bregman-regularized" is a misnomer: $T_{\text{reg}} = (1 - \lambda) T + \lambda \Pi_K(T)$ contains no Bregman divergence, and is in fact the Krasnoselskii-Mann averaged form	U2	VERIFIED	L1433, L2313: $T_{\text{reg}}(K) = (1 - \lambda) T(K) + \lambda \Pi_K(T(K))$ with the Moreau projection invoked (L1443); the KM identification confirmed by inspection of the formula - this is GLM's most constructive T5 contribution
Prop 16.2 / Prop 16.7 items 3-4: " $\kappa_V(R_{\text{max}}) = \kappa_V(R_i) = 0$ verified to 10^{-9} " is $0 = 0$ on an object where κ_V is undefined; "falsifiable prediction" is a category error	U2 U6	VERIFIED	Prop 16.2 (L2424-2434): "agrees within machine precision (10^{-9})"; Prop 16.7 (L2593-2600): "Both sides are zero to within 10^{-9} " under the heading "Falsifiable prediction". A discrete RAF set carries no declared geometry for Prop 4.4's curvature
The inverse-limit / colimit confusion recurs (section titled "Filtered-Colimit Construction" whose proposition item is labeled "Inverse limit" with projection arrows)	U2	VERIFIED	Prop 16.7 item 2 (L2588-2590): "Inverse limit: ... The 15 projection arrows R_{max} to R_i ... witnessing the universal property of the limit" inside Sec 16
Def 19.9's persistent-homology "contractibility test" tests homology, not contractibility; Remark 17.5's Phase III is a 30% statistics-matching tolerance	U2	VERIFIED	Def 19.9 (L4306-4315): barcodes via ripser; Remark 17.5 (L2749-2760): "mean, max, and min within relative tolerance $\tau = 0.30$ ". Persistent homology cannot detect contractibility; GLM's Poincare/averaging replacement (D6) absorbs the one real insight (limit cycles are not endpoint failures)
Table 1 residuals are English phrases ("residual = Holevo information ..."), not fiber objects	U2	VERIFIED	L1232-1234: "residual = Holevo information $I(\rho_{\text{out}}; \hat{\rho}_{\text{in}})$ "; the seven residuals are semantics, not typed objects
Sections 8 and 15 duplicate the contraction analysis	U2	VERIFIED	Both state the product bound (L1417 vs L1499) and both carry Banach-contraction narrative; Sec 8 is analytic, Sec 15 numerical (375-configuration grid) - partial duplication, full merge feasible
The "kernel constant κ of order unity" in $C_{\text{fat}} = \sqrt{\nu/2} \times \kappa$ is never derived	U2	VERIFIED	L1978, L7246: the constant is declared "of order unity" rather than derived from the Levy first-passage normalization, which is explicit

Table 4. Type-level and structural defects.

II.4 Coherence and editorial layer (all persisting in v21)

Claim	Audit	Verdict	Evidence
Six different network counts persist	U1 U2	VERIFIED	v21 tex: "four real biochemical networks" (L135, L329), "two" (L426, L4285), "ten" (L10166); the 2nd-wave count ($2/4/2/4/10$) persists in redistributed form - a single network registry is still missing
Table 4 is claimed updated three times but lists only the nine E1-E9 studies	U1 U2	VERIFIED	Caption (L4101): "Nine novelty-assessment elevation studies"; update claims at L4663, L4787, L5094 (Remarks 19.18, 19.22, 19.28)
"No open conjectures remain" contradicts two explicitly open items in the body	U1 U2	VERIFIED	L7475: "no open conjectures remain"; L7229: "five conjectures ... now closed"; vs Remark 20.2's own "Conjecture 21.1 ... separate, well-defined open problem" and Remark 7.8's open endofunctor construction
Section 19 is a change log (19.1-19.23), and version-history framing persists in definition titles	U1	VERIFIED	Subsections 19.1-19.23 span L4053-L6659; Def 3.21's title carries "v10 main definition"; "backward comparability with the v1-v9 elevation studies" (L787)
Session provenance persists: "the user", audit names, commit hashes, folder paths, COLOMBOS notes	U1	VERIFIED	Counts in the v21 PDF: "the user" x6 (case-sensitive; 12 case-insensitive), "Qwen" x34, commit 07e6d85 x5, COLOMBOS x3, patch-script names and download/ paths throughout Sec 19
The authorship statement is contradicted by the body's round-by-round narration	U1	VERIFIED	L7539-7543: "the model did not originate novel [results]" vs Sec 19's documented AI-executed study lineage (v1-v21 rounds, audit responses, user-requested variants)

Claim	Audit	Verdict	Evidence
Def 2.1's "not a curvature 2-form" disclaimer is contradicted by Cor 4.14 / E1 treating the radial identity as curvature content	U2	VERIFIED	Def 2.1 (L204-216) disclaims; Cor 4.14 computes H_{geo} from the same radial object; E1 (Sec 10) uses " $\kappa_V = a^2$ " by Stokes theorem"
Claims A-E are simulations of textbook identities or of the simulator's own generating law, not experiments	U2	VERIFIED	Table 3 (L2055-2062): "slope = 1", "arev = 1.0" - identities of the generating law; GLM's reclassification as estimator calibration is sound and consistent with the 2nd-wave finding
The zero-control and partial-correlation honesty layer is intact	this assessment	VERIFIED	E24 (L6310-6340): zero-control MWU $p = 7.8 \times 10^{-22}$; partial $r = +0.251$ reported with its own limits ("not unconfounded"); E26/E27 protein-layer nulls and magnitude parity reported without inflation

Table 5. Coherence and editorial layer.

II.5 Items the audits list that v21 already resolved

The audits reviewed the 131-page v20. The v21 mechanical-integrity round (commit ddbb384, 29 verified corrections) resolved a subset of overlapping items before this assessment: the broken citation key orth2011comprehensive now renders correctly (resolving one of DeepSeek's four citation-integrity items; the other three - twelve never-cited references, six false "Cited in SX" annotations, and the authorship statement - remain, still HELD as P0-editorial pending the strategic decision); the E25 +0.187 (n=241) confluences were corrected to the primary +0.191 with an explicit common-set reconciliation note; the Def 3.18 step labels and the Table 6 clipping were repaired (P1); and the EC-number corrections were applied. Nothing else in the six audits' scope was touched by v21, so every remaining verdict above is current. The practical consequence for planning: of the combined 2nd-wave-plus-3rd-wave defect inventory, only the mechanical stratum is complete; the editorial, mathematical, and structural strata are exactly as the audits describe them.

PART III

Audit Errors: Corrections and Refutations

The standing instruction for this assessment - and the reason it exists - is that audit claims are not taken at face value. Eight audit statements failed verification or need material qualification. None of them changes the direction of the overall verdict; several of them change what v2 should implement, which is precisely why they are worth isolating before any repair work starts.

#	Audit	Erroneous or imprecise statement	Correction, with basis
1	Kimi (U4)	R2 as stated: "for y to x , $\kappa_t / \ t - T1\ ^2$ converges to the curvature form along the trajectory tangent"	The limit is the Hessian-metric ENERGY, not curvature: $D_{\text{phi}}(v(T1), v(t))$ is approximately $(1/2) v \cdot \dot{T1} ^2 t - T1 ^2$ under the surrogate metric, i.e. the squared norm of the tangent vector (gpt's "local response metric" and opus's "Phi itself"), not a second-order curvature contraction. A one-parameter path cannot yield sectional curvature without second differences. With R2 corrected, kimi's framework agrees with gpt/opus instead of conflicting with them
2	Muse (U5)	"The mask is the indicator that $\text{iota}(v)$ and $\text{iota}(v_{\text{wt}})$ lie in different strata of B - a discrete proxy for crossing the singular locus"	Imprecise for the manuscript's actual mask. The mask $1[\Delta_b > \tau b_{\text{wt}}]$ selects nonzero single-knockout BIOMASS DEFICIT (opus's reading, exactly right for Def 3.21); a non-essential knockout can still reroute fluxes and cross active-set strata with mask = 0 - the manuscript's own $dR(g) = \{r : v_r(\text{KO}) - v_r(\text{WT}) > 10^{-6}\}$ is defined precisely for such changes. The stratum-crossing reading requires the nondegeneracy assumption "no rerouting without deficit". The defensible formulations are GLM's (mask = gene-level positive part, consistent with $[\cdot]_+$ in κ_V 's own definition) and opus's (deficit sparsification); kimi's Lambda-restriction is a redefinition that changes the formula

#	Audit	Erroneous or imprecise statement	Correction, with basis
3	Opus (U6)	"Def 3.21 ... is the mixed-difference curvature with knockouts as perturbations and biomass deficit as the deficit function ... precisely the standard FBA measure of flux rerouting capacity"	Aspirational, not descriptive. Def 3.21 as written is a single-knockout squared displacement (a Φ -slot value in opus's own notation); the epistasis object requires the double-knockout recomputation of Route C2. The reinterpretation is the correct TARGET for v2; presenting it as the current content of Def 3.21 would reproduce at the audit layer exactly the conflation the audits are trying to remove
4	GLM (U2)	"for the prototype instantiation, product of $\alpha_i = 0.697 \times 1.15 = 0.802$ on the enlarged box $X = [0, 1.15]^d$, which removes the need for the projection patch"	Not derivable from the manuscript as stated: $f_2 = 1.15 \times$ does not map $[0, 1.15]^d$ into itself ($1.15 \times 1.15 > 1.15$), so the enlarged-box route needs a different construction than the one implied. The SOUND part of GLM T5 is the Krasnoselskii-Mann identification, which this assessment confirms structurally: $T_{\text{reg}} = (1 - \lambda) T + \lambda \Pi_K(T)$ already IS the KM averaged form. The 0.802 constant should be treated as to-be-derived, not as fact
5	GLM (U2)	Lemma 4.10 repair bound " $ \text{grad } r \leq 2 \beta L_d (\text{diam } K + D)$ "	The identity $\text{grad } r = \beta \sum \pi_j \text{grad } q_j$ is verified correct (the outer $-\tau$ cancels). The constant needs max over j of sup over K of $[d - D]_+$ rather than $\text{diam } K + D$, unless codewords are assumed to lie in K - codewords are arbitrary strings. A one-line completion, stated here so v2 does not import an off-by-geometry constant
6	GLM (U2)	" κ_V^{FBA} (Def 2.6 ...)" in the thread table	Def 2.6 is the RAF-set definition; the FBA statistic is Def 3.21. Citation slip only - GLM's claim set otherwise had the highest verification rate of the six audits
7	DeepSeek (U1)	"a broken citation key" (present tense, in the citation-integrity list)	STALE: resolved in v21 (commit ddbb384). DeepSeek's other three citation-integrity items remain valid. A v20-era artifact, not an error of judgment
8	GLM (U2)	"Report that the fitted exponent lies outside the quoted CI rather than 'verified to within 1.4%'"	Direction reversed: it is the THEORY value 1.500 that lies outside the fitted CI $[1.471, 1.493]$, and the manuscript already discloses this once with a bias explanation (L2013-2017). The genuine defect is the inconsistent framing - three "verified to within 1.4%" sites vs one honest CI site - and the un-derived kernel constant (which GLM correctly flags separately)

Table 6. Audit errors and qualifications found by this assessment.

Two systemic observations complete this part. First, no audit made an error in the direction that would soften the manuscript's problems: every correction above either tightens an already-confirmed defect or shifts a repair detail. The audits are individually reliable and jointly complementary, which is what makes the Part V synthesis worth building. Second, the corrections cluster where theory meets the manuscript's actual formulas: kimi's R2 and muse's mask reading both fail against the literal text of Def 3.21 and E22, which is a structural argument for opus's sequencing advice - do the mechanical re-typing pass FIRST, because several "broken" objects turn out to be mislabeled rather than wrong, and several "unified" claims turn out to be conflations rather than theorems; only a typed naming layer makes the difference visible.

PART IV

Cross-Audit Triangulation

Treating the six audits as independent measurements of the same object makes their agreements informative in a way any single audit cannot be. Seven triangulation points emerged; five are strong multi-audit convergences, and two are the genuine disagreements that Part V has to adjudicate.

Convergence	Audits	Content and strength
C1. Typed renaming with the geometric object as the only kappa	6 / 6	Every audit ends with a distinct typed name for each object ($\kappa^{\text{geom}} / K^{\text{FBA}} / \kappa^{\text{hat}^{\text{emp}}} / \Delta^{\text{R}} / \kappa^{\text{Lambda}2} - \kappa^{\text{ko}} - \kappa^{\text{t}}$). Muse's migration checklist, kimi's "no bare kappa" rule, and gpt's naming table are enforceable by mechanical search - the only kind of rule that survives contact with a 10,830-line source

Convergence	Audits	Content and strength
C2. The FBA active-set / mpLP critical-region stratification is the bridge substrate	5 / 6 (U2 U3 U4 U5 U6)	GLM T7(a) (mpLP critical regions ARE the base strata), opus G-B (curvature lives on basis-change loci; LP value function piecewise-linear so curvature is singular on strata), gpt's mapping table (active LP basis to stratum), kimi's embedding (knockout simplex with face restrictions), muse's stratum-crossing reading. This is the strongest mathematical convergence of the wave: the geometric thread's strata and the FBA thread's active-set decomposition are the same object, and opus's corollary - in the generic non-degenerate case the smooth curvature vanishes and ALL curvature concentrates on the transition loci - explains in one sentence why the essentiality mask, the non-abelian signatures, and the constraint-switch hysteresis all live in the same place
C3. The provable glue is the energy/response-divergence level	4 / 6 (U3 U5 U6 + U4 after correction)	Gpt's proposition ($D_{\phi}(v_{\beta}(p + h \xi), v_{\beta}(p)) = (h^2/2) \text{Hessian-quadratic} + O(h^3)$), muse's Theorem U first display, opus's Phi-level reading, and kimi's R2 (once corrected to energy) all state the SAME theorem in different notation: the FBA and time-course statistics are consistent finite-difference proxies for the induced Hessian-metric energy, not for curvature. This theorem is provable now and is the honest content of the current E24-E27
C4. Curvature measurement needs two directions	2 / 6 explicit (U3 U6), all 6 implicit	Gpt's four-condition plaque ($p, p + h \xi, p + k \eta, p + h \xi + k \eta$) with $(P - I)/(hk)$ converging to the curvature operator; opus's Route C1 (second differences on existing time courses - free) and Route C2 (double knockouts - the epistasis design). The theory's discriminating prediction is which PAIRS show super-additive deficit
C5. The shared repair list	all repairs have 3+ audit sponsors	Cor 4.14, Lemma 4.10, Thm 4.11(e), Remark 20.2, E13 Thm A, E13 Thm B, Cor 17.3, the realization-functor fork (define monoidically on concrete optics - GLM D3, opus's evaluation-at-fiber-map, kimi's residual-assigned optics - or excise thread C), the tuple freeze, the KM retargeting of Thm 8.2, and one notation table. Where 3+ audits state a repair for the same defect, the repairs coincide in substance
C6. Strong-form unification is not provable in scope	6 / 6	Kimi's warning is the crispest ("strong-form unification ... probably not provable within this manuscript's scope and should not be claimed"; R4 is conditional with H3 a paper of its own); muse's Theorem U is explicitly conditional with a fallback table; gpt's discrete-realization theorem is conditional on a transport construction "not supplied by FBA automatically"; opus requires "an explicit inequality or error term" or the paper splits; GLM prices T2 and T7 at months each; DeepSeek demands "all internal inconsistencies reconciled" before any such claim. The joint position: only conditional transfer theorems with labeled hypotheses are honest
C7. The real disagreement: delete or demote	U2 vs U1	GLM would delete Section 17 wholesale, Cor 4.14, Theorem B, Prop 16.2, Prop 19.7, and the Zeno/Levy analogies; DeepSeek's standing instruction is "prioritize rigorous elevation of math, repair and demoting to conjecture over removal". Part V resolves this with a three-tier adjudication

Table 7. Triangulation across the six audits.

One convergence deserves emphasis because it is invisible in any single audit. GLM's T2 (non-abelian information concentrates at the boundary as a group-valued jump cocycle), opus's G-B corollary (curvature is singular, supported on basis-change loci), kimi's embedding (the discrete statistic is a sampled connection coefficient on a masked subbundle), and muse's mask reading (crossing the singular locus where Gamma jumps) are four descriptions of ONE phenomenon: in a stratified system, the second-order object that the experiments can see is concentrated on the strata transitions, not distributed over the smooth cells. That statement is simultaneously the paper's actual thesis (opus: "currently invisible"), the reason the essentiality mask is not an ad hoc device, and the reason the non-abelian signature appears exactly at constraint switches. A v2 that states this once, early, and derives everything else from it would be a different and much stronger paper than one that renames three objects and stops.

PART V

The Synthesized Unification Architecture

The six proposals are not competitors; they are layers of one architecture that none of them states completely. What follows is the adjudicated assembly - the product of this assessment's strengthening mandate. Each layer names its source audits, what it fixes, and what it costs.

Layer 0 - Notation protocol (mechanical, v2 pass 1)

Adopt muse's migration checklist as written, kimi's "no bare kappa" rule (the symbol kappa appears only as the family or with an instantiation decoration), and gpt's naming table: κ^{geom} for the geometric curvature, $K^{\text{FBA}} / \Delta^{\text{FBA}}$ for the knockout statistic, $\Phi_{\text{obs}} / \Delta^{\text{time}}$ for the trajectory statistic, with forbidden-use columns per muse's disambiguation table (never called curvature, never compared to E, never claimed to be Prop 4.4's object). A one-page notation-generation table in Section 2 enforces the rule mechanically. Fix the typed-citation defects in the same pass: Def 6.1 cites the D_V /curvature datum rather than "the κ_V of Definition 2.1"; dist_D is applied only to strings (the three $D_\phi(\text{dist}_D(R), \text{dist}_D(R0))$ sites become dist_D applied to declared encodings, or are replaced by the Layer-1 datum); the SAVGS tuple is frozen as (B, E, P, epsilon, Gamma) with Thm 3.14's (Theta, pi, Delta) recorded once as the special case. Opus's caution applies: run this pass BEFORE any theorem repair, and expect it to expose mislabelled-but-correct proofs and correct-but-mislabelled claims - the pass itself is the diagnostic.

Layer 1 - The central object, factored (the adjudication)

Five candidate central objects were proposed. The adjudication: adopt opus's curvature datum as the combinatorial DEFINITION, keep kimi's slot family as the smooth organization of the geometric thread, keep gpt's realization datum as the formal wrapper for the response-divergence layer, and keep GLM's h-margin version as the viability-theoretic instantiation that carries the erosion inequality. The rationale is in the table below; the one-sentence version is that opus's mixed difference is the only candidate under which the RAF instantiation is well-typed (it fixes Prop 16.2 for free), Prop 4.4 is recovered as a theorem rather than an assumption (Proposition G), and the FBA object becomes epistasis - the honest target - at the cost of one recomputation.

Candidate	Type safety	RAF instantiation	Recovers Prop 4.4?	Recompute cost	Verdict
Opus: $\kappa = \sup \text{ over } a, b \text{ of } [\Phi(abx) - \Phi(ax) - \Phi(bx) + \Phi(x)]_+$	Clean (set-level)	Well-typed via (X, A, Φ) with catalyst removals	Yes, as $\epsilon^2 \times$ Hessian coefficient (Proposition G)	Route C2 recompute for FBA	ADOPT as the definition
Kimi: slot family $\kappa[d, c, \Lambda]$	Clean on smooth strata	Needs a declared geometry	Yes, as the infinitesimal slot value	None for existing studies	ADOPT as the smooth organization (Lambda-domains; R1/R3 usable)
GPT: realized divergence Δ_R	Most complete typing (mask, normalization, observation map explicit)	Out of scope by design	As Cur_x of Δ_R	None for existing studies	ADOPT as the formal wrapper for response divergences
GLM: $\kappa_V = \sup \max_\alpha [-D h_\alpha]_+ / h_\alpha$ (h a smooth margin)	Clean	Out of scope	Direct (it IS the viability form)	None	ADOPT as the viability instantiation; carries T3 erosion inequality

Candidate	Type safety	RAF instantiation	Recovers Prop 4.4?	Recompute cost	Verdict
Muse: canonical $\kappa^{\text{geom}} = (1/E) \sup [\text{Breg}_r] +$	Clean but keeps the algorithmic surrogate inside the geometric object	Undefined	It IS Prop 4.4	None	PARTIAL - keep as the envelope-linked form; E-normalization noted as a choice, not a necessity

Table 8. The five candidate central objects and the adjudication.

The essentiality mask is re-instantiated in this layer, and the five audit readings collapse to two that are correct: the mask is the gene-level positive part on the single-knockout deficit (GLM's consistency reading and opus's sparsification reading, which coincide for Def 3.21 as written), and kimi's Lambda-restriction is available as a variant that must be declared as a redefinition if adopted. Muse's stratum-crossing reading survives only as a conditional statement under a stated nondegeneracy assumption. GLM's pre-registration framing converts the v10 promotion episode from a defect into a declared estimator choice only if the $\tau = 0.05$ scale is stated ex ante in v2 - one sentence, but it is the difference between definition shopping and an estimator protocol.

Layer 2 - The two provable transfer theorems (now)

Two transfer statements are provable with the material already in hand, and they are the honest glue for the existing studies. T-energy (gpt's proposition, = muse's Theorem U part 1, = kimi's R2 corrected): on a fixed active-set stratum with the regularized optimum v_{β} (strictly concave objective, unique and differentiable by construction - gpt's v_{β} or kimi's barrier, with the strict-convexity caveat of Part III item routed through the regularized optimum rather than through the code surrogate), the FBA and time-course statistics are consistent finite-difference proxies for the induced Hessian-metric energy. T-bound (kimi R1 + R3, both "trivial after definition" but the first true sentences connecting the flagship to the abstract): plane-restricted and domain-restricted statistics are bounded by the full-sup object once the embedding is fixed, and the one-direction knockout sample is a singleton-plane instance. What must NOT be claimed is the strong form (all six audits agree; kimi's R4 stays conditional with H1-H3 labeled, muse's fallback table is the printed escape hatch).

Layer 3 - The three new measurements (strengthened from the audits)

This is where the assessment completes the audits' weaker suggestions into a concrete work program. M1 (opus Route C1, cost: days): compute genuine second differences $\kappa\text{-hat}(t) = [\Phi(t + 2 \Delta) - 2 \Phi(t + \Delta) + \Phi(t)] +$ on the EXISTING time courses - data availability verified: GSE64021 provides six points per condition (15 min to 6 h), the M3D carbon-exhaustion series provides a reference plus four stationary points, and the E10/E22 dynamic-FBA trajectories already export κ per time point T1-T8. Opus calls this a two-line computation; the assessment confirms the data exist and adds that it converts E24-E27 from "validates the deficit function" to "measures the deficit function AND its first curvature slice". M2 (gpt's plaque, cost: weeks): the four-condition estimator with regularized optima, directions chosen from the high- κ genes, converging to κ^{geo} under gpt's stated hypotheses - with gpt's own caveat printed in the theorem statement: the GC parallel transport must be constructed from FBA sensitivity data; FBA does not supply it automatically. M3 (opus Route C2, cost: weeks): sampled double-gene knockouts in iJO1366 - all pairs within the top κ decile plus stratified random pairs - measuring the epistasis excess directly. The assessment's augmentation: M3 is executable in silico now (cobrapy plus quadratic regularization, the same tooling as E12/E15/E16), which converts opus's "the experiment the theory actually predicts" from an aspiration into a runnable study; the theory predicts WHICH PAIRS show super-additive deficit, and that prediction is testable against the mixed-difference ranking. Together M1-M3 move the empirical program from "a conjectured transfer" to "a measured transfer at the energy layer plus first curvature-slice measurements" - the single largest value-add this wave can deliver.

Layer 4 - The theorem repair tiers (resolving delete-vs-demote)

The C7 disagreement resolves into three tiers, satisfying DeepSeek's elevation preference wherever the object is recoverable and GLM's coherence requirement wherever it is not. Tier R (repair in place): Lemma 4.10 by the softmax

identity (verified in Part II, with the Part III constant completion); Thm 4.11(e) by the rational grid plus equicontinuity - to which this assessment adds the missing second half: the direction supremum is closed by a countable dense subset (the supremum of a continuous function over a compact set equals its supremum over any dense subset), so both continua in (20) are dischargeable by two short lemmas; Remark 20.2 by GLM's three-line corollary (on $\{h \geq \epsilon\}$: $\kappa_V = [-Dh]/h \leq \text{Lip}(h)/h \leq \text{Lip}_{\text{loc}}(E)/\epsilon$ - dimensionally correct, provable, and the honest content of "the upper-bound language is a theorem"); Prop 3.16 by the one-line radius-2 fix; Remark 2.4 by the stabilizer relabel; Thm 3.5 by GLM's T2 jump-cocycle restatement (the correct theorem is BETTER than the stated one: smooth curvature at ϵ^2 , non-abelian information as a boundary jump cocycle indexed by the crossing word); Thm 3.4 by the exact gauge form with the BCH form demoted to a derived corollary; E13 Thm A by the r-in-S line plus Tarski/Hordijk-Steel; Thm 8.2 by the KM retargeting with "Bregman-regularized" renamed; Thm 3.14 by the tuple freeze. Tier D (demote with absorption): E13 Thm B to an adjoint-uniqueness conjecture (kimi's restatement); Section 17 (HoTT layer) to a companion note, with its one real insight - limit cycles are not endpoint failures - absorbed into the closure criterion via Poincare-section sampling (GLM D6), and Remark 17.5's 30% tolerance retired with it; the Zeno/Levy material re-labeled as analogy or attached to kimi's κ^l fluctuations; Claims A-E reclassified as estimator calibration; Prop 16.2 and 16.7(3-4) re-typed through the Layer-1 datum; Prop 19.7 replaced by the at-most-|U| stabilization statement; Cor 4.14 to a worked appendix example with the normalization explicit. Tier E (excise): the six false "Cited in SX" annotations and twelve never-cited references (the HELD P0-editorial items, now scheduled); "no open conjectures remain"; the version-history framing; the network-count registry replaced by one table. Nothing in Tier E destroys recoverable mathematics - which is the test DeepSeek's policy demands and GLM's list, as amended, now passes.

PART VI

Implementation Plan and the v2 Protocol

The versioning protocol (binding)

Per the project directive issued with this wave: the latest manuscript version is not to be touched. v21 (commit ddbb384) is hereby the FROZEN REFERENCE BASELINE - the state of the manuscript against which every audit verdict above was rendered, and the artifact on which the strategic decision is to be made. From here on, every major rewrite is a NEW document: `journal_manuscript_v2.tex` is created from the v21 source and carries the Layer 0-4 program; subsequent rounds produce v3, v4, and so on, each as a new file with its own compile, QA, and commit. Even minor errata are not applied to the frozen v21; if an erratum matters it belongs in v2. The frozen baseline guarantees that the audits, this assessment, and the eventual strategic decision all refer to one immutable object - the failure mode this protocol prevents is the one the manuscript itself documents, where v10 definitions, v14b corrections, and v17 results coexist in one text with no clean reference point.

Sequencing

Step	Action	Prerequisite	Cost	Output
0	Strategic decision: single coherent article (GLM's 60-90 pp estimate; DeepSeek's single-narrative requirement) vs. split into a main paper plus companion notes (opus/gpt's excision path; the HoTT and dynamics material as companions)	this assessment (the decision input is now complete)	days of deliberation	A written decision document; all downstream scoping depends on it
1	v2 pass 1 - Layer 0 mechanical re-typing of the full source (renaming, typed citations, tuple freeze, network registry, version-framing strip)	step 0	1-2 weeks	<code>journal_manuscript_v2.tex</code> ; expect 1-2 additional results killed and at least one "broken" proof revealed as fine (opus's prediction, endorsed)
2	Layer 2 theorems (T-energy, T-bound) written and proven in v2	step 1	2-4 weeks	The first honest flagship-abstract link

Step	Action	Prerequisite	Cost	Output
3	M1 second differences on existing time courses (E28-class study)	step 1 only	days	New results files + v2 subsection; the first measured curvature slice
4	M2 plaquette estimator + M3 double-KO epistasis study (E29/E30-class)	steps 1-2	2-6 weeks	The mixed-difference ranking and its test against super-additive deficit pairs
5	Layer 4 Tier R theorem repairs (the nine repair-in-place items)	step 1	3-8 weeks (T2 and the envelope grid are the long poles)	Repaired theorem statements with short proofs or honest "conjecture" labels
6	Tier D demotions and absorptions (Section 17 companion, Thm B conjecture, Claims A-E reclassification, Prop 16.2 re-typing, Cor 4.14 appendix)	step 5	1-3 weeks	A shorter main paper plus companion notes
7	Tier E excisions: P0-editorial citation integrity (6 false annotations, 12 uncited refs), "no open conjectures" claim, authorship statement rewrite	step 0 (strategic decision), per the HELD status	days	Citation-integer manuscript; the HELD P0-editorial items are released for execution here, coordinated with P3 provenance excision
8	Full re-audit of v2 (external, same protocol as this wave) before any submission decision	steps 1-7	external	A v2 audit wave; iterate

Table 9. Implementation sequence. Total to a coherent single article: GLM's 6-12 month estimate is consistent with this table at research pace; the mechanical and measurement strata (steps 1-4) are a 1-2 month core.

Risks and fallbacks

Three risks dominate. R1: the Layer-1 embedding may fail on inspection - kimi's own fallback ("if R3's embedding fails, fall back to the renaming-only option") and muse's disambiguation table are the printed escape hatches, and the Layer 2 theorems survive either way, so the downside is bounded. R2: T2 and T7 are genuinely new mathematics (GLM's pricing: 1-3 months and months respectively) - the plan treats them as the paper's contribution rather than as prerequisites; if they stall, the conditional-transfer version is publishable with them stated as conjectures, which is exactly the epistemic status all six audits demand. R3: scope creep back into v21 - prevented by the versioning protocol, which makes the frozen baseline a hard boundary. One scheduling note: M1 (step 3) has no prerequisites beyond the frozen artifacts and the Layer 0 names, and could run immediately after step 1 to give the strategic decision (step 0) an empirical data point it currently lacks: whether the measured curvature slice correlates with anything. That result would inform both the single-article and the split options, which is why this assessment recommends starting M1 as soon as pass 1 lands, in parallel with the step-0 deliberation.

PART VII

Reliability Scorecard and Joint Verdict

Audit	Checkable claims	Verified	Corrected / stale	Distinctive contribution
DeepSeek (U1)	14	13	1 stale (citation key)	The editorial specification; the delete-vs-demote policy that shaped Tier D/E
GLM (U2)	41	39	2 corrections (0.802 arithmetic; Def 2.6 slip) + 1 direction fix (Levy CI)	The structural backbone: D1-D6, T1-T7, the deletion list, and both genuinely new theorems (T2 jump-cocycle, T7 mpLP bridge); highest verification rate
GPT (U3)	15	15	0	The most type-careful reading: realization datum, plaquette estimator, the transport caveat, "biomass deficit is not a metric"

Audit	Checkable claims	Verified	Corrected / stale	Distinctive contribution
Kimi (U4)	12	10	1 correction (R2 energy vs curvature) + 1 completion (strict convexity)	The constructive mathematics: R1/R3 (first provable link), the embedding, the strong-form warning
Muse (U5)	19	18	1 correction (mask reading)	The migration checklist and Theorem U - the only drop-in implementation text
Opus (U6)	17	16	1 correction (mixed-difference as description)	The empirical re-reading: Phi-not-kappa, Routes C1/C2, "curvature is singular on the strata" - the paper's invisible thesis made visible

Table 10. Reliability scorecard. "Checkable claims" counts statements this assessment could test against the source; totals: 118 claims, 111 verified outright, 8 carried corrections or stale status.

Joint verdict

The six audits are unanimous on the diagnosis and complementary on the cure. The diagnosis, fully verified here: the manuscript's central weakness is not length, provenance, or citation hygiene - it is that one name currently denotes three mathematical objects of different type, so that the abstract's proposition and the empirical flagship are about different things; and the theorem layer that should carry the weight contains seven load-bearing statements that are false, vacuous, incoherent, or proven for the wrong object. The cure, triangulated five ways: a typed notation layer; a factored central object of which the geometric curvature, the knockout statistic, and the trajectory deficit are declared instantiations; an energy-level transfer theorem that is provable now; curvature-level measurements that are computable now; and a tiered repair-demote-excise program that elevates everything recoverable and absorbs the rest. The single decision that gates all of it - one article or a main paper plus companions - is now made on a clean basis: a frozen, mechanically-verified v21, a complete defect inventory across two audit waves, and a synthesized architecture with its first steps already specified to the line level.

This assessment makes no edits to the manuscript, per the directive: v21 is frozen, and everything specified in Parts V and VI is work for journal_manuscript_v2 and later versions. The immediate next actions, in order: take the strategic decision; create v2 and run the Layer 0 pass; start M1 in parallel; then proceed down Table 9.